

Fluid Pressure Outline

①

① Pressure Equations and Units

$$\rho = ? \quad \gamma = ?$$

$$p = \rho gh \quad \text{or} \quad p = \gamma h \quad (\text{where, } \gamma = \rho g)$$

ρ = mass density

γ = Specific Weight = $\rho \cdot g$

Fresh Water:

$$\gamma = 62.4 \text{ lb/ft}^3$$

$$\gamma = 9810 \text{ N/m}^3$$

$$\rho_w = 1000 \text{ kg/m}^3 \times 9.81$$

$$= 9810$$

② Pressure Rules/Principles

- P increases w/ depth

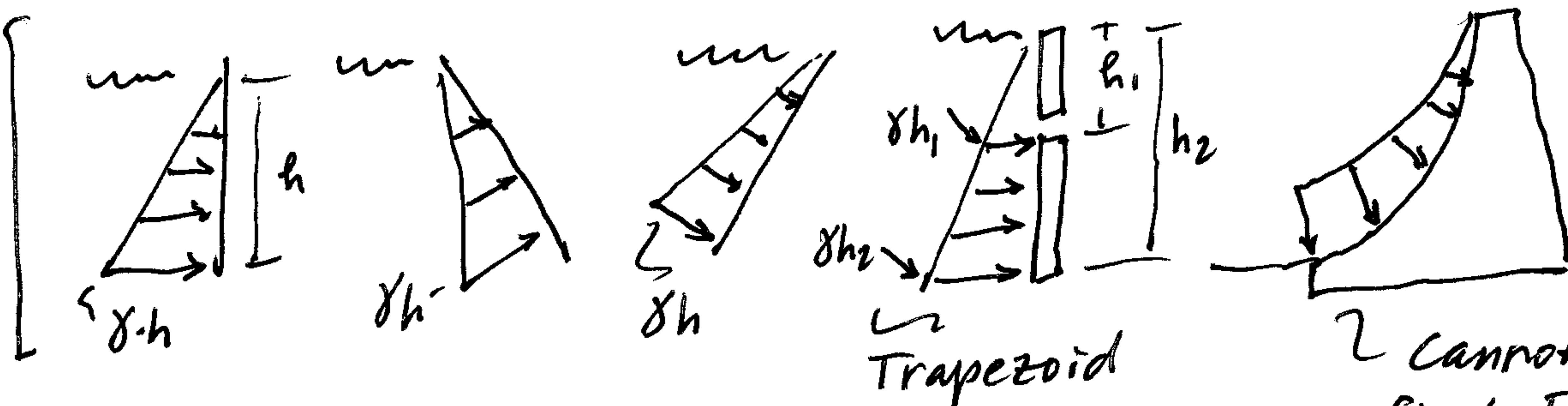
- Swimmers

- Submarines

- Concrete Forms ($\gamma_{con} \approx 150 \text{ lb/ft}^3$)!

- P acts $\perp$ to any surface

A
Flat Plates



- We seek $\vec{F}_R$ and its location.

F_R = Volume of Pressure Distribution

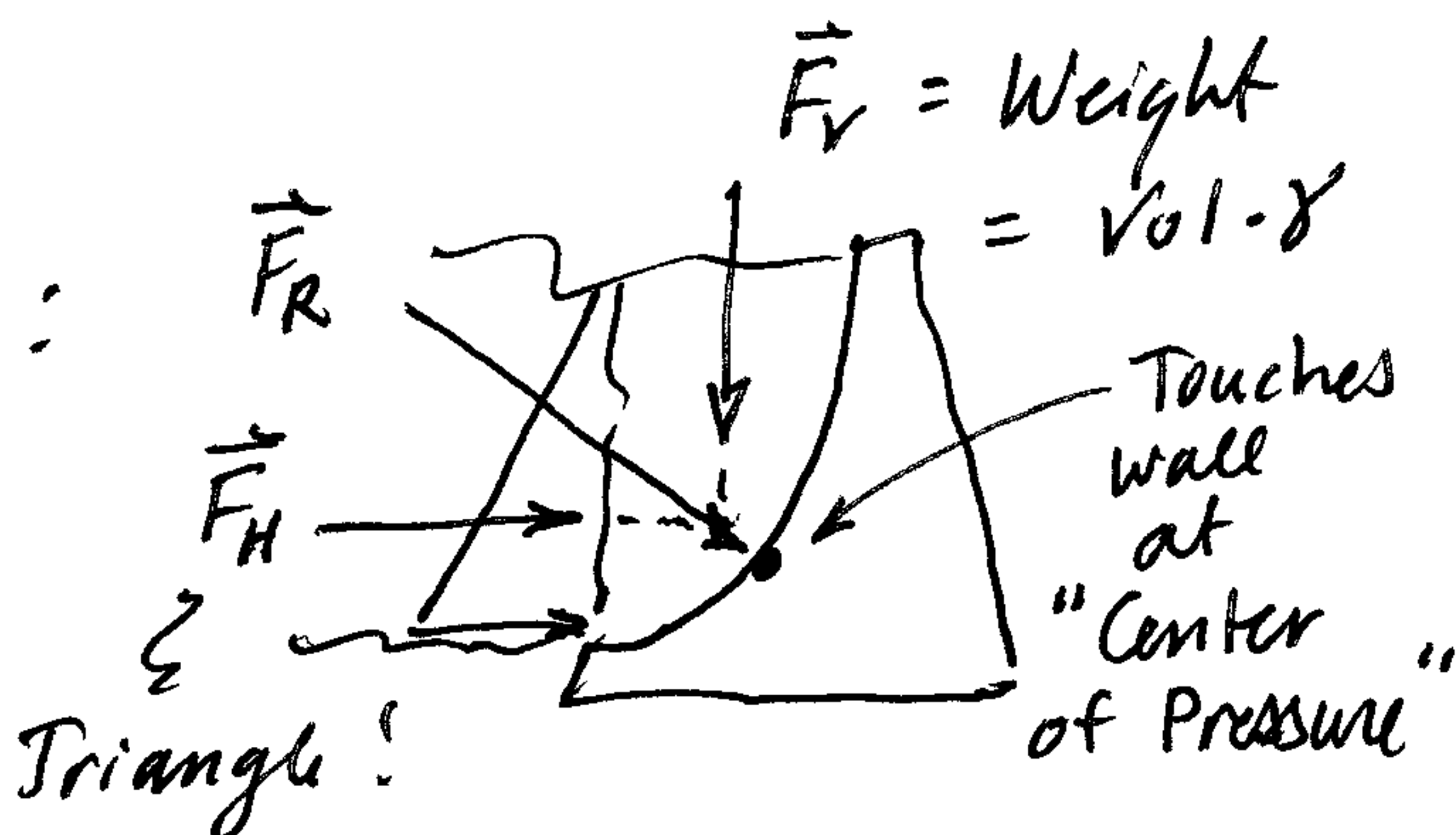
$$F_R = \int_A P \cdot dA$$

(Most general fluid equation...)

B
Curved Surfaces

- For curved surfaces: (constant width)

$$\vec{F}_R = \vec{F}_H + \vec{F}_V$$



One more equation: "GaZA" ...

or

$$F_R = GZA$$

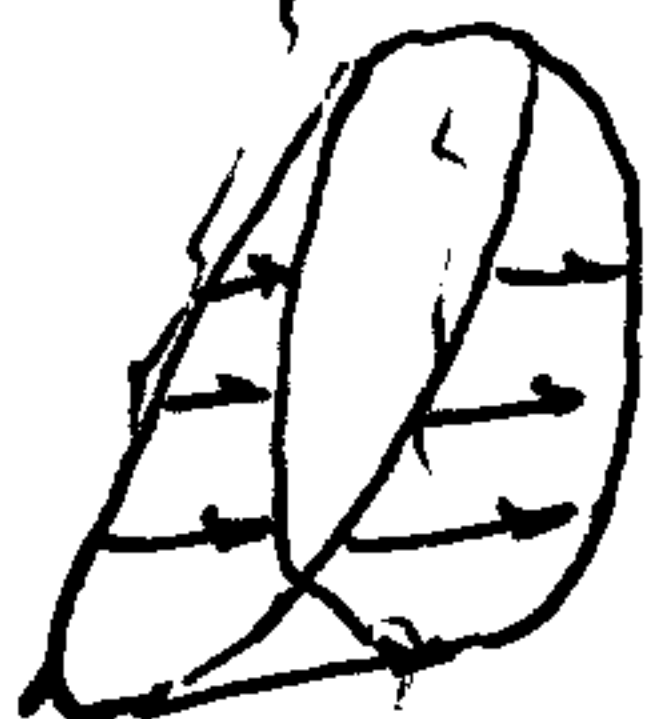
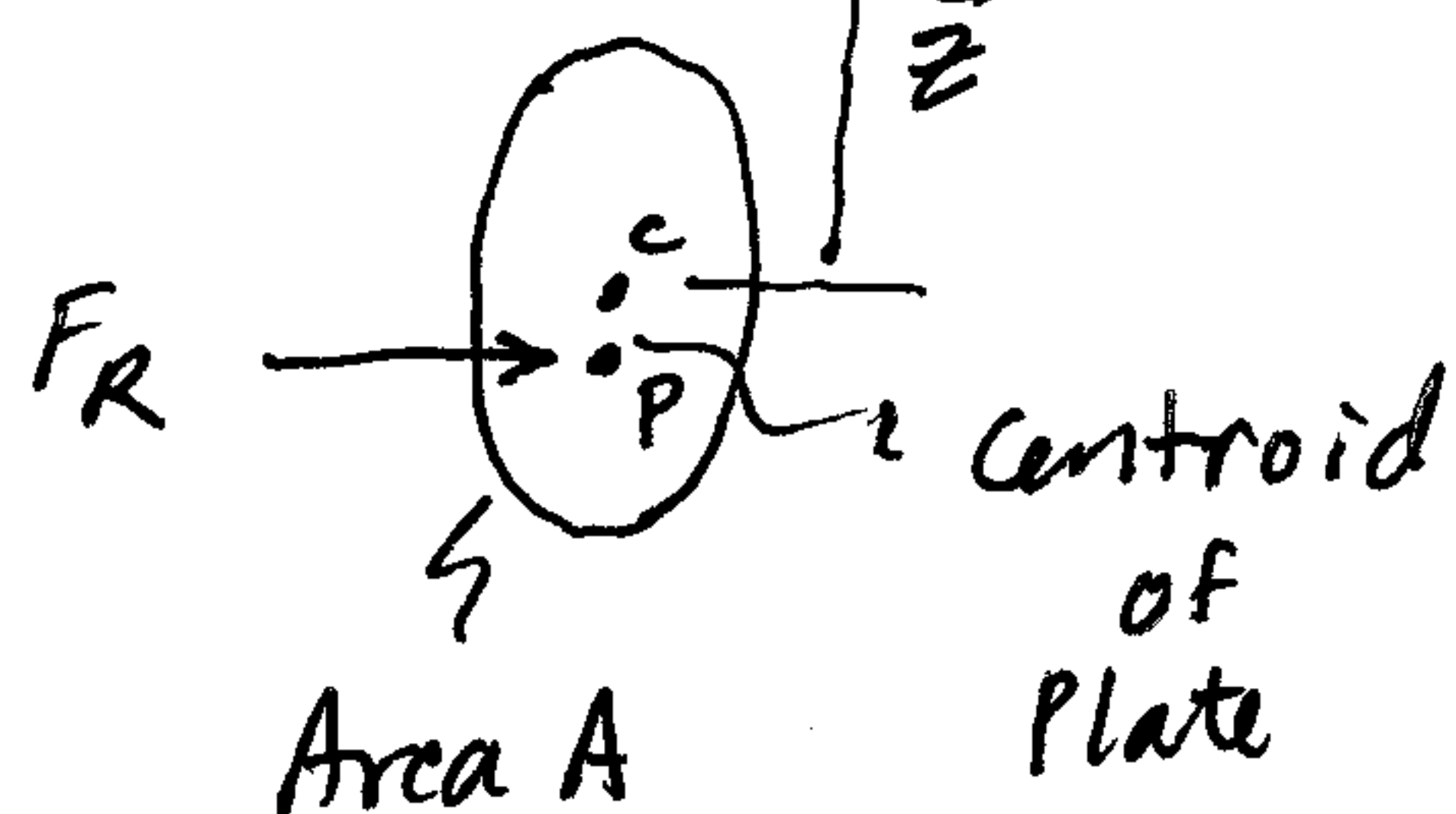
$$F_R = \gamma \cdot \bar{z} \cdot A$$

"Gamma · $\bar{z}$ · Area"

Use for

Flat Plates w/ Non-constant width

①



Pressure Distrib
increases w/
depth ...
Cannot easily
find F_R ,
exapt ...

$$F_R = \gamma \cdot \bar{z} \cdot \text{Area}$$

Important!

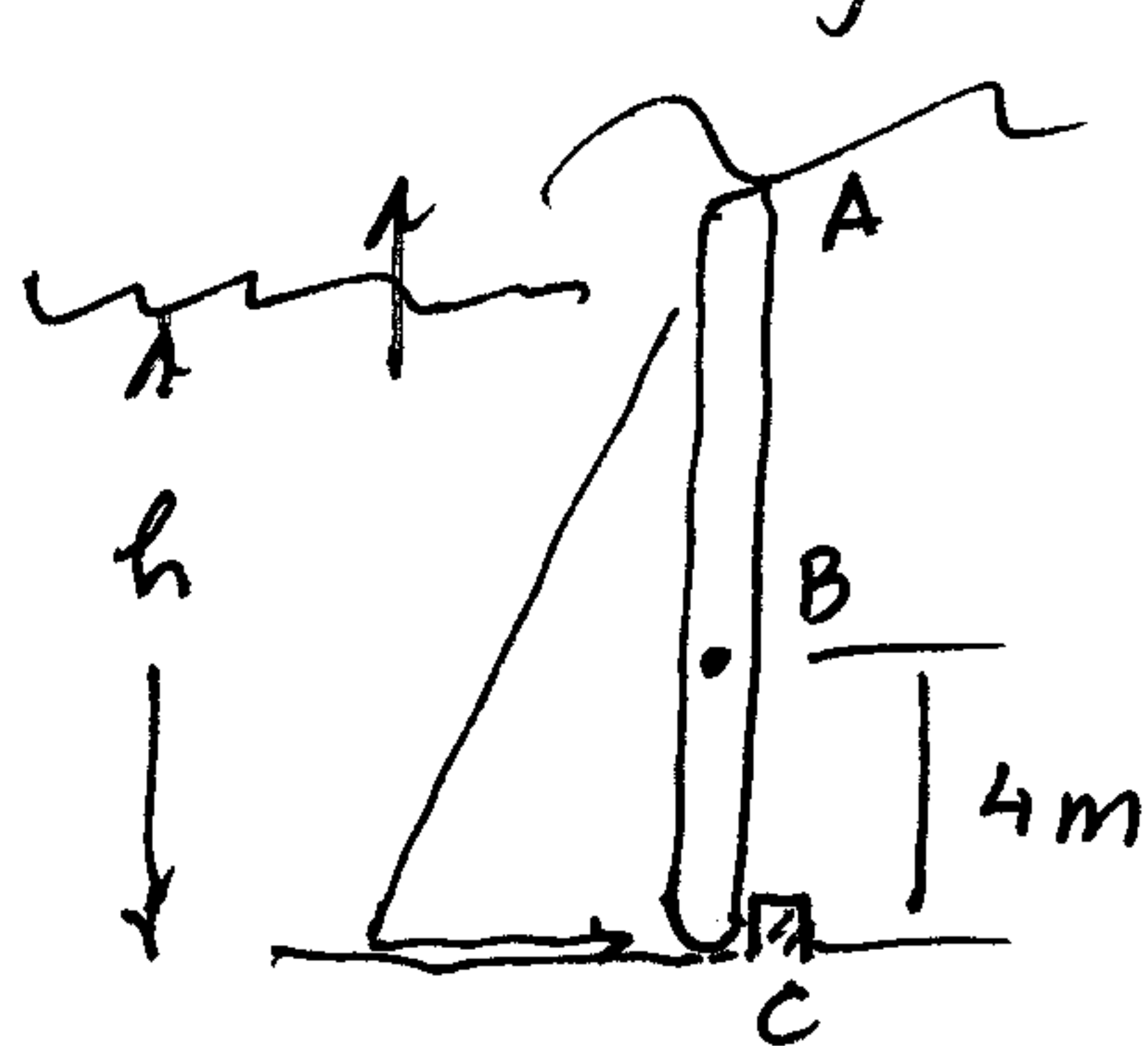
$\bar{z}$ evaluated at
centroid of Area A (at C)

BUT, F_R acts at
the Center of Pressure, (at P!)

Flat Plates w/ Non-constant Width:

② Can also integrate. Ugh!
See example problems.

Gate Opening Problems

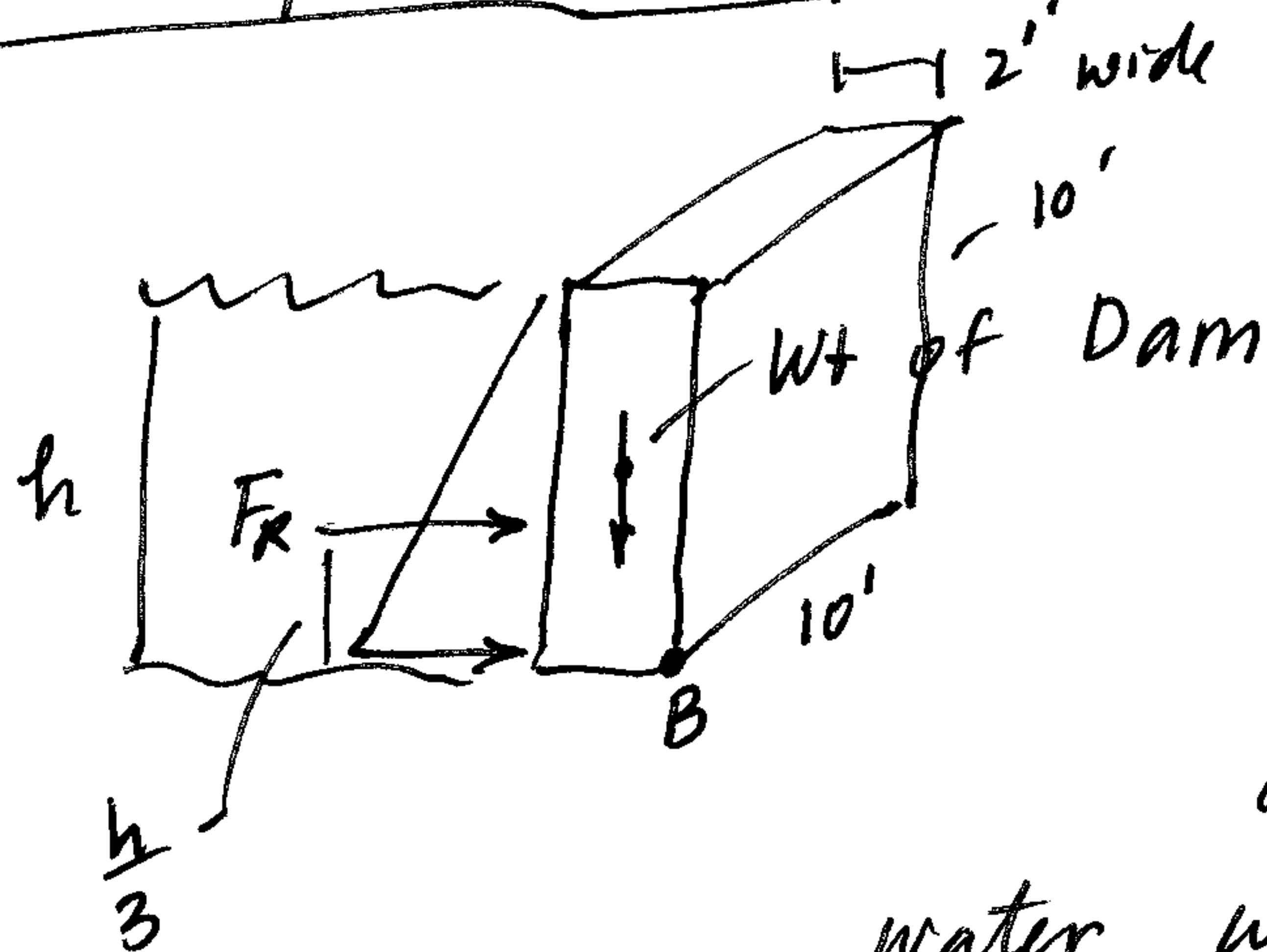


Rising water level...

at what height does gate ABC open?

Gate ABC can pivot around pin at B.

Gravity Dam Tipping Problem



Concrete Dam:

$$W = 150 \frac{\text{lb}}{\text{ft}^3} (2)(10)(10) = 30,000 \text{ lb}$$

at what depth of water will this dam tip?

(Neglect friction sliding case)

$$F_R = \frac{1}{2} b h \cdot W = \frac{1}{2} (\gamma h) (h) (10 \text{ ft}) \left. \begin{array}{l} \text{Acts at } h/3 \\ P = \text{base} \end{array} \right\} = 5 \gamma \cdot h^2$$

$$\gamma = 62.4 \frac{\text{lb}}{\text{ft}^3}$$

$$+ \sum M_B = 0;$$

$$F_R \cdot (h/3) - 30,000 (1) = 0$$

$$(5 \gamma \cdot h^2) (h/3) = 30,000$$

$$\frac{5}{3} \gamma \cdot h^3 = 30,000$$

$$h^3 = 288.5$$

$$\boxed{h = 6.61 \text{ ft}}$$